The group router with a round-robin strategy ensures that each photo processor receives its fair share of messages.

6.3.4 Consistent hashing strategy

The previous section showed that routers are an easy way to scale up and out. But there can be a problem with sending messages to different routees. For example, consider the following common scenario.

A manager splits work between two workers in parallel. One type of worker obfuscates sensitive data, while the other type enriches the data with third-party vendors. When these two actors finish their tasks, they have to send the product to a final actor that aggregates both pieces of work; let’s call it the aggregator. However, the message from the data obfuscator and the data enricher can be sent to the wrong aggregator, as figure 6.6 shows. The problem is ensuring that the manager’s original message, which was split between the workers, ends up being merged back into the same aggregator. It would not be surprising if workers sent their finished work from the same message to different aggregators if you used the strategies seen so far. In figure 6.6, aggregator Y may not be able to merge the two parts—or, worse, it may merge them incorrectly, and the error may go unnoticed.